hblibPresentation: User Manual

for hblibPresentation.sty v20261109T0615

Haluk Bingol

Faculty of Computer and Information Sciences
Yeditepe University

2026-11-14

Outline

1. Getting started
2. Text and emphasis
3. Sections and separators
4. Extra material
5. Tables, algorithms, keys
6. Code on slides
7. References
8. Notes and troubleshooting

Getting started

Loading, packages, theme

Loading the libraries

```
\documentclass[aspectratio=169,10pt,compress]{beamer}

\usepackage{../../hblib/hblibPresentation}     % first: the base library
\usepackage{../../hblib/hblibCodeoutput}       % code and output read from files
\usepackage{../../hblib/hblibDatastructures}   % arrays, stacks, queues, trees

\title[Arrays 1D: Lecture]{One-Dimensional Arrays}
\subtitle{\cisOOO\ -- Week 7 lecture}
\author[Bingol]{Haluk Bingol}
\institute[FCIS]{Faculty of Computer and Information Sciences\\ Yeditepe University}
\date{{\tiny \hbTimeStamp}}
```

- hblibPresentation loads hblibCommon from its own folder: keep the hb libraries in one folder (`hblib/`)
- Load with `\usepackage`, without `.sty`; a relative path gives a harmless "You have requested package ..." warning
- Use `[t,fragile]` frames: code listings and `\lstinline` need `fragile`
- Compile twice: the outline and "i / N" need the second run

The hb libraries

Library	Contents
hblibCommon	shared by slides and articles: fonts, packages, algorithms, colors and emphasis, code styles, `javacode`, `\hCodeJ`, `\hCode`, time stamps, `\reffig` ..., `\hAbs`, column type `L`; loaded by the two below, not by the document
hblibPresentation	slides: theme, `\hSection`, `\hSubsection`, `\hSeparator`, `\hSectionSkip`
hblibArticle	articles (lab descriptions, exams): page, titles, contents, section numbering, footer, links
hblibDatastructures	optional: arrays, stacks, queues, strings, grids, trees, commit graphs
hblibCodeoutput	optional: code and output read from files, in boxes

- This manual covers hblibPresentation *and* the hblibCommon commands a deck uses most; the full list of hblibCommon is in hblibArticle-UserManual
- In the tables, (C) marks a command of hblibCommon: it works in articles too

What the library loads

Package	Gives
by hblibCommon	
fontenc (T1), lmodern	accented (Turkish) letters, `< > |`; sharp fonts
amsmath, amssymb	math, `\lvert` (`\hAbs`), symbols
graphicx, xcolor	`\includegraphics` (also .eps), colors
tikz	drawings; libraries positioning, matrix, arrows.meta, calc
calc, array, booktabs	`\widthof`; column type `L{width}`, `\toprule` ...
forest, datetime2, menukeys	trees; `\today` as 2026-11-14; `\menu`, `\keys`
listings	program code: styles codestyle, codesnippet ..., `javacode`
algorithm2e	pseudo code: ruled, vlined, linesnumbered
by hblibPresentation	
fp	`\FPeval`
enumitem	only outside beamer (it breaks beamer's lists)

Not needed any more, so not loaded: inputenc (UTF-8 is the default), xparse (in LaTeX since 2020), epstopdf (graphicx converts .eps by itself).

Theme and frame numbers

The library sets the look of every deck:

```
\useoutertheme{split}           % sections and subsections in the head line
\useinnertheme{rounded}         % rounded blocks and bullets
\usecolortheme{whale}
\usecolortheme{orchid}
\setbeamertemplate{navigation symbols}{}                        % no navigation symbols
\setbeamertemplate{page number in head/foot}[totalframenumber]  % "i / N" at the bottom
```

- A deck can change it after `\usepackage{hblibPresentation}`, e.g. `\usecolortheme{beaver}`
- The short title `\title[short]{...}` and the short author `\author[Bingol]{...}` go to the foot line

Text and emphasis

Colors, inline code, small helpers

Emphasis

Write	Result	Use
`\hDefined{reference}`	reference	a term defined here
`\hEmph{important}`	important	an important word
`\hRemark{macOS}`	macOS	a side remark
`\hQuestion{Why?}`	**Q.** Why?	a question to the audience
`n\hIndex{2}`	n^2	a small raised index
`\hReference{CLRS}`	[CLRS]	a source, small
`\hCite{p. 12}`	[p. 12]	a source, tiny
`\hUrl{oracle.com}`	[oracle.com]	a web address
`\hColorBox{x}`	x	old name of `\hEmph`
`\win`, `\mac`	**Windows**, **macOS**	the two systems

All (C). Colors used: red, colorHEmph, orange, colorIndex, colorStop, colorGo.

Inline code, sizes, names, dates

```
\hCode{int[] a = new int[5];}    \hCode{project\_dir}    \hCode{\{3, 1, 4\}}
\hFontLarger{2}{2 pt larger}  \hFontLarger{-2}{2 pt smaller}
\hAbs{x - y}    \cisOOO    \cisOOT    \hbTimeStamp    \hbTimeStampDraft
```

`int[] a = new int[5];` `project_dir` `{3, 1, 4}`

2 pt larger 2 pt smaller

$|x - y|$ CIS111 CIS112 2026-11-14 –Draft– v2026-11-14/10:57:37

- `\hCode`: typewriter, colored (colorHCode), 2 pt smaller. Write `\_ \{ \}` as in text. It also works in frame titles and in `\hSection` titles (the PDF bookmark gets plain text)
- A deck may change the color: `\definecolor{colorHCode}{RGB}{0,90,200}`
- All of these are (C), from hblibCommon
- For syntax-colored code use `\hCodeJ` (also hblibCommon; the frame must be fragile)

Sections and separators

`\hSection`, `\hSubsection`, `\hSeparator`

\hSection: a section and its title frame

```
\hSection[short]{Long}{line}          % between frames, not inside one
\hSection[Basics]{Creating and using an array}{Create it, visit it with a loop}
\hSection[End]{Summary}
```

- `short` (optional): in the head line; the outline and the frame show `Long`
- `Long`: the title on the frame, `\Huge`, orange
- `line` (optional): under the title, `\Large`
- The previous frame was made by `\hSection[Sections]{Sections and separators}{...}`
- Old form `\hSection[short]{Long}[line]` is no longer accepted: write `{line}`
- `\hSectionSkip` as the line marks a section for self-study (next frames)

\hSubsection, \hSeparator, \hSectionSkip

```
\hSubsection{small line}{Title}   % \subsection plus a title frame (small line on top)
\hSeparator{Title}{line}          % a title frame, without a section
\hSection[Extra]{Extra material}{\hSectionSkip}
```

The next three frames are made by these lines, with `\hSubsection{Sections}{A subsection}`, `\hSeparator{A separator}{without a section}` and `\hSection[Extra]{Extra material}{\hSectionSkip}`.

Sections

A subsection

A separator

without a section

Extra material

(skip this section, read at home)

An optional part: switches

A deck can make optional parts with a switch, as the lecture decks do:

```
\newif\ifActivities  \Activitiestrue    % in the preamble; \Activitiesfalse hides the frames

\ifActivities % EXTRA activities V
  \begin{frame}[t,fragile] \frametitle{Quick check}
    ...
  \end{frame}
\fi % EXTRA activities A
```

This is plain LaTeX (`\newif`), not part of the library; it combines well with `\hSection[...]{...}{\hSectionSkip}`.

Tables, algorithms, keys

Column type L, algorithm2e, menukeys

Tables: column type L{width}

```
\begin{table}
  \footnotesize
  \begin{tabular}{@{}L{0.30\textwidth}L{0.55\textwidth}
      @{}}
    \toprule
    \textbf{Term} & \textbf{Meaning} \\
    \midrule
    element & one stored value \\
    index   & its position: $0, 1, \dots, n-1$;
              long text wraps in the column \\
    \bottomrule
  \end{tabular}
\end{table}
```

Term	**Meaning**
element	one stored value
index	its position: $0, 1, \dots, n-1$; long text wraps in the column

`L{w}` is a paragraph column, ragged right; `\textwidth` here is the column's width.

Algorithms (algorithm2e), step by step

```
\begin{algorithm}[H]
  \caption{Largest($a$)}
  \KwIn{an array $a$ of $n \ge 1$ numbers}
  \KwOut{the index of the largest}
  $b \gets 0$\;
  \For{$i \gets 1$ \KwTo $n - 1$}{
    \uncover<2->{\lIf{$a[i] > a[b]$}{$b \gets i$}}
  }
  \uncover<3->{\Return{$b$}\;}
\end{algorithm}
```

Algorithm 1: Largest(a)

Input: an array a of $n \geq 1$ numbers
Output: the index of the largest
1 $b \leftarrow 0$
2 **for** $i \leftarrow 1$ **to** $n - 1$ **do**

- The number "Algorithm 1" stays the same on all three slides (`\resetcounteronoverlays{algocf}`)
- `\Fn{name}{body}`: "function ... end"; captions in `\small`
- `\DontPrintSemicolon` in a deck's preamble hides the semicolons

Algorithms (algorithm2e), step by step

```
\begin{algorithm}[H]
  \caption{Largest($a$)}
  \KwIn{an array $a$ of $n \ge 1$ numbers}
  \KwOut{the index of the largest}
  $b \gets 0$\;
  \For{$i \gets 1$ \KwTo $n - 1$}{
    \uncover<2->{\lIf{$a[i] > a[b]$}{$b \gets i$}}
  }
  \uncover<3->{\Return{$b$}\;}
\end{algorithm}
```

Algorithm 1: Largest(a)

Input: an array a of $n \geq 1$ numbers
Output: the index of the largest

```
1 b ← 0
2 for i ← 1 to n − 1 do
3     if a[i] > a[b] then b ← i
```

- The number "Algorithm 1" stays the same on all three slides (`\resetcounteronoverlays{algocf}`)
- `\Fn{name}{body}`: "function ... end"; captions in `\small`
- `\DontPrintSemicolon` in a deck's preamble hides the semicolons

Algorithms (algorithm2e), step by step

```
\begin{algorithm}[H]
  \caption{Largest($a$)}
  \KwIn{an array $a$ of $n \ge 1$ numbers}
  \KwOut{the index of the largest}
  $b \gets 0$\;
  \For{$i \gets 1$ \KwTo $n - 1$}{
    \uncover<2->{\lIf{$a[i] > a[b]$}{$b \gets i$}}
  }
  \uncover<3->{\Return{$b$}\;}
\end{algorithm}
```

Algorithm 1: Largest(a)

Input: an array a of $n \geq 1$ numbers
Output: the index of the largest

1 $b \leftarrow 0$
2 **for** $i \leftarrow 1$ **to** $n - 1$ **do**
3 ⌊ **if** $a[i] > a[b]$ **then** $b \leftarrow i$
4 **return** b

- The number "Algorithm 1" stays the same on all three slides (`\resetcounteronoverlays{algocf}`)
- `\Fn{name}{body}`: "function ... end"; captions in `\small`
- `\DontPrintSemicolon` in a deck's preamble hides the semicolons

Menus and keys (menukeys)

```
\menu{Run > Debug As > Java Application}    \keys{Ctrl + Shift + B}    \keys{\shift + \cmd + B}    \keys{F
    6}
```

Run › Debug As › Java Application Ctrl + Shift + B ⇧ + ⌘ + B F6

- menukeys defines `\cmd` (the Mac command key ⌘), so the library uses `\hCode` for commands
- In a table: **Windows** Ctrl + C **macOS** ⌘ + C

Code on slides

lstlisting, javacode; updating old decks

Code typed in a frame

```
\begin{lstlisting}[language=Java]
if (n > 0) {
    n--;          // 4 spaces of indentation
}
\end{lstlisting}

\begin{javacode}
if (n > 0) {
    n--;          // 4 spaces of indentation
}
\end{javacode}
```

```
1 if (n > 0) {
2     n--;          // 4 spaces of indentation
3 }
```

```
if (n > 0) {                                    java
    n--;          // 4 spaces of indentation
}
```

- A plain lstlisting uses `codestyle` of hblibCommon: colored, line numbers, as in articles
- `javacode`, `bashcode`, `style=codesnippet`: framed, no numbers, a language tag (javacode)
- The frame must be `[fragile]`; code and output from files: hblibCodeoutput

References

`\cite` and a .bib file

Citing and the references frame

```
Arrays~\cite{TheJavaTutorialsArray}; queues and stacks~\cite{cormen2022introduction}.

  \begin{frame}[t,fragile] \frametitle{References}
  \setbeamertemplate{bibliography item}[text]      % [1], [2] instead of icons
  \bibliographystyle{IEEEtran}
  \tiny\bibliography{../../hblib/hblibDataStructures}
\end{frame}
```

Result: Arrays [**?**]; queues and stacks [**?**] (the last frame of this manual).

- Compile: `pdflatex`, `bibtex`, `pdflatex`, `pdflatex`
- The file is `hblibDataStructures.bib` (capital S), next to the libraries
- Only cited entries are listed; `\nocite{*}` lists all

Notes and troubleshooting

Troubleshooting

Message or problem	Cause	Fix
Missing \begin{document} after loading	`\input` of the .sty	`\usepackage`
You have requested package '../../hblib/...'	a relative path	harmless
"i / ??" or an empty outline	first run	compile again
Undefined control sequence in a listing frame	frame not `fragile`	`\begin{frame}[t,fragile]`
the line of an old `\hSection[s]{L}[line]` shows as stray text	old form	`\hSection[s]{L}{line}`
a `{...}` group right after `\hSection` becomes its line	the optional `{line}`	put something else first, e.g. `\relax`
indices purple, not blue	hblibDatastructures loaded after	as intended (it sets colorIndex)
File '../../hblib/hblibCommon.sty' not found	hblibCommon missing	all hb libraries in one folder
Command \hEmph already defined	old and new libraries mixed	update all hb libraries together
Undefined control sequence \hCodeOutput; style mystyle undefined	removed in v20261109T0615	see "Updating an old deck"
Citation '...' undefined, "[?]"	bibtex not run	run `bibtex`, then pdflatex twice
enumitem options ignored	not loaded in beamer	beamer's own lists

Quick reference

Command	Result
`\hSection[short]{Long}{line}`	section and title frame
`\hSubsection{small line}{Title}`	subsection and title frame
`\hSeparator{Title}{line}`	title frame only
`\hSectionSkip`	"(skip this section, read at home)"
`\hDefined \hEmph \hRemark \hQuestion`	emphasis ({text})
`\hIndex \hReference \hCite \hUrl`	small notes ({text})
`\hCode{text}`, `\hFontLarger{N}{text}`	inline code; size change
`\hAbs{x}`	$\|x\|$
`\cisOOO \cisOOT \win \mac`	CIS111, CIS112, **Windows**, **macOS**
`\hbTimeStamp \hbTimeStampDraft`	date stamps
`L{width}`	table column
colors: colorHEmph colorStop colorGo colorIndex colorHCode gitred	`\definecolor` to change
from hblibCommon: all rows from `\hDefined` to `L{width}`, and the colors	

References